

Arindam Bose

PH.D. • RADAR SIGNAL PROCESSING RESEARCHER

Washington DC, United States

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Summary

Radar Systems Engineer with specialized expertise in imaging radar systems for drone detection and tracking applications. Proven experience in designing, developing, and optimizing radar signal processing algorithms for high-resolution imaging, target classification, and clutter suppression in complex environments. Strong background in MIMO radar, beamforming, Doppler processing, and real-time embedded implementation. Adept at designing and integrating radar hardware with advanced signal processing pipelines to enhance situational awareness in airborne and ground-based surveillance systems. Committed to advancing state-of-the-art drone detection technologies through innovative, reliable, and scalable radar solutions.

Professional Experience

KMB Telematics Inc.

Washington DC, USA

SENIOR RESEARCH ENGINEER, RADAR SIGNAL PROCESSING

Oct. 2020 – Present

RESEARCH INTERN, RADAR SIGNAL PROCESSING

Summer 2019, 2020

- Developing high-resolution imaging radar capable of detecting and tracking small UAVs, including low-flying or stationary drones, with on-the-move operation to locate launch sites and pursue threats.
- Building a full-stack radar signal processing toolchain, including digital system and FPGA firmware, along with radar data visualization and UI software using Python, Verilog, Rust, and C.

University of Illinois at Chicago

Chicago, IL, USA

RESEARCH ASSISTANT, WAVEOPT LAB, DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

May 2016 – Dec. 2020

- Designed and implemented multiple efficient algorithms for transmit waveform synthesis in MIMO radar systems, enhancing overall performance and adaptability.
- Collaborated closely with Prof. M. Soltanalian on research in radar signal processing and optimization theory, contributing to ongoing studies and the development of my Ph.D. thesis.

TEACHING ASSISTANT, DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Aug. 2015 – May 2020

- Courses assisted: Digital signal processing, Statistical signal processing, Image analysis and computer vision.

Mitsubishi Electric Research Laboratories

Cambridge, MA, USA

RESEARCH INTERN, SIGNAL PROCESSING GROUP

Summer 2018

- Developed innovative techniques and efficient imaging algorithms for terahertz (THz) time-domain spectroscopy systems.

Cognizant Technology Solutions Pvt. Ltd.

Kolkata, India

PROGRAMMER ANALYST, HEALTH CARE PRACTICE

Apr. 2013 – Jul. 2014

- Designed, developed, and maintained multiple Java-based web applications, tailored to meet diverse client requirements and deliver scalable, high-performance solutions.
- Designed and implemented dynamic web services and sophisticated web pages using JSP, HTML, CSS, and JavaScript, ensuring responsive design and seamless user experience.

Education

University of Illinois at Chicago

Chicago, IL, USA

PH.D. IN ELECTRICAL ENGINEERING

2021

M.S. IN ELECTRICAL ENGINEERING

2020

- Ph.D. Thesis: *Waveform Synthesis for Active Sensing with Emerging Applications*
- Committee: Prof. Mojtaba Soltanalian, Prof. Dan Schonfeld, Prof. Daniela Tuninetti, Prof. Rashid Ansari, Dr. Perry Wang

West Bengal University of Technology

Kolkata, India

B.TECH. IN ELECTRONICS AND COMMUNICATION ENGINEERING

2012

Selected Publications

JOURNAL ARTICLES

- J5 A. Bose, B. Tang, W. Huang, M. Soltanalian, and J. Li, **Waveform Design for Mutual Interference Mitigation in Automotive Radar**, *arXiv preprint arXiv:2208.04398*. Aug. 2022
- J4 A. Bose, B. Tang, M. Soltanalian, and J. Li, **Mutual Interference Mitigation for Multiple Connected Automotive Radar Systems**, in *IEEE Transactions on Vehicular Technology*, vol. 70, no. 10. Oct. 2021

- J3 **A. Bose, S. Khobahi, and M. Soltanalian, Efficient Waveform Covariance Matrix Design and Antenna Selection for MIMO Radar**, in *Signal Processing*, vol. 183. Jun. 2021
- J2 **A. Ameri, A. Bose, J. Li, and M. Soltanalian, One-Bit Radar Processing With Time-Varying Sampling Thresholds**, in *IEEE Transactions on Signal Processing*, vol. 67, no. 20. Oct. 2019
- J1 **A. Bose, M. Soltanalian, Constructing Binary Sequences With Good Correlation Properties: An Efficient Analytical-Computational Interplay**, in *IEEE Transactions on Signal Processing*, vol. 66, no. 11. Jun. 2018

CONFERENCE PRESENTATIONS

- C13 **Z. Esmaeilbeig, A. Bose, and M. Soltanalian, Ambiguity Function Shaping in FMCW Automotive Radar**, *IEEE Asilomar Conference on Signals, Systems, and Computers 2023*, Pacific Grove, CA. Nov. 2023
- C12 **Z. Esmaeilbeig, A. Bose, and M. Soltanalian, Mutual Interference Mitigation in PMCW Automotive Radar**, *IEEE European Microwave Week 2023*, Berlin, Germany. Sep. 2023
- C11 **S. Khobahi, A. Bose, and M. Soltanalian, Deep One-Bit Compressive Autoencoding**, *IEEE Statistical Signal Processing Workshop 2021*, Rio de Janeiro, Brazil. Jul. 2021
- C10 **A. Bose, A. Ghauri, and M. Soltanalian, Limits of Transmit Beamforming for Massive MIMO Radar**, *IEEE Asilomar Conference on Signals, Systems, and Computers 2020*, Pacific Grove, CA. Nov. 2020
- C9 **S. Khobahi, A. Bose, and M. Soltanalian, Deep Radar Waveform Design for Efficient Automotive Radar Sensing**, *IEEE Sensor Array and Multichannel Signal Processing Workshop 2020*, Hangzhou, China. Jun. 2020
- C8 **A. Bose, S. Khobahi, and M. Soltanalian, Joint Optimization of Waveform Covariance Matrix and Antenna Selection for MIMO Radar**, *IEEE Asilomar Conference on Signals, Systems, and Computers 2019*, Pacific Grove, CA. Nov. 2019
- C7 **A. Bose, A. Ameri, and M. Soltanalian, Waveform Design for One-Bit Radar Systems Under Uncertain Interference Statistics**, *IEEE Asilomar Conference on Signals, Systems, and Computers 2019*, Pacific Grove, CA. Nov. 2019
- C6 **P. Wang, T. Koike-Akino, A. Bose, R. Ma, P. Orlik, W. Tsujita, K. Sadamoto, H. Tsutada, and M. Soltanalian, Learning-Based Shadow Mitigation for Terahertz Multi-Layer Imaging**, *IEEE International Conference on Infrared, Millimeter, and Terahertz Waves 2019*, Paris, France. Sep. 2019
- C5 **A. Bose, A. Kadu, H. Mansour, P. Wang, P. Boufounos, P. Orlik, and M. Soltanalian, THz Multi-Layer Imaging Via Nonlinear Inverse Scattering**, *IEEE International Conference on Infrared, Millimeter, and Terahertz Waves 2019*, Paris, France. Sep. 2019
- C4 **I. A. Arriaga-Trejo, A. Bose, A. G. Orozco-Lugo, and M. Soltanalian, Design of Unimodular Sequence Sets with Good Correlation and Complementary Correlation Properties**, *IEEE Global Conference on Signal and Information Processing 2018*, Anaheim, CA. Nov. 2018
- C3 **A. Bose, I. A. Arriaga-Trejo, A. G. Orozco-Lugo, and M. Soltanalian, Generalized Cyclic Algorithms for Designing Unimodular Sequence Sets with Good (Complementary) Correlation Properties**, *IEEE Sensor Array and Multichannel Signal Processing Workshop 2018*, Sheffield, UK. Jul. 2018
- C2 **A. Bose, N. Mohammadi and M. Soltanalian, Designing Signals with Good Correlation and Distribution Properties**, *IEEE International Conference on Acoustics, Speech and Signal Processing 2018*, Calgary, Alberta, Canada. Apr. 2018
- C1 **A. Bose and M. Soltanalian, Efficient Construction of Polyphase Sequences With Optimal Peak Sidelobe Level Growth**, *IEEE Global Conference on Signal and Information Processing 2017*, Montreal, Canada. Nov. 2017

BOOK CHAPTERS

- B1 **A. Bose, J. Li, and M. Soltanalian, One-Bit Cognitive Radar**, Booktitle: *Next Generation Cognitive Radar Systems*, Editors: K. V. Mishra, B. Shankar, and M. Rangaswamy, IET Press. 2023

PATENTS

- P1 **P. Wang, T.-K. Akino, P. Orlik, A. Bose, Learning-Based See-Through Sensing Suitable for Factory Automation**, *US Patent and Trademark Office*, Patent ID: US20210064013A1. 2019

Honors & Awards

- Signal Processing Society Chicago Chapter Appreciation**, IEEE, Chicago, IL, USA 2019
- Associate of the Month**, Cognizant Technology Solutions, Kolkata, India 2014
- Winner**, The Telegraph Knowhow Innovation Hub, INFOCOM 10-11, Kolkata, India 2011
- Special Prize**, Science and Engineering Fair, Kolkata, India 2010 – 2013
- Educational Scholarship**, Central Government of India, Kolkata, India 2008 – 2012